

MEDIATEK

MT8676 Yocto eCall

Version: 1.0
Release date: 2024-08-12

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版本记录

版本	日期	作者	描述
1.0	2024-08-12	刘宇田	正式版本

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1 eCall

1.1 概述

1.1.1 简单介绍

eCall 是一种车载道路安全系统，可在发生严重事故时自动呼叫紧急服务。一旦 eCall 传感器检测到车辆受到严重撞击或手动发起呼叫，eCall 车载系统 (IVS) 就会与相关公共安全应答点 (PSAP) 建立 e112 语音连接。通过语音连接向 PSAP 发送一组最小数据集 (MSD)，其中包括准确的地理位置数据。

2015 年 4 月，欧洲议会规定，自 2018 年 3 月 31 日起，所有新车型都必须配备 eCall 技术。eCall 系统概览如图 1-1 所示。

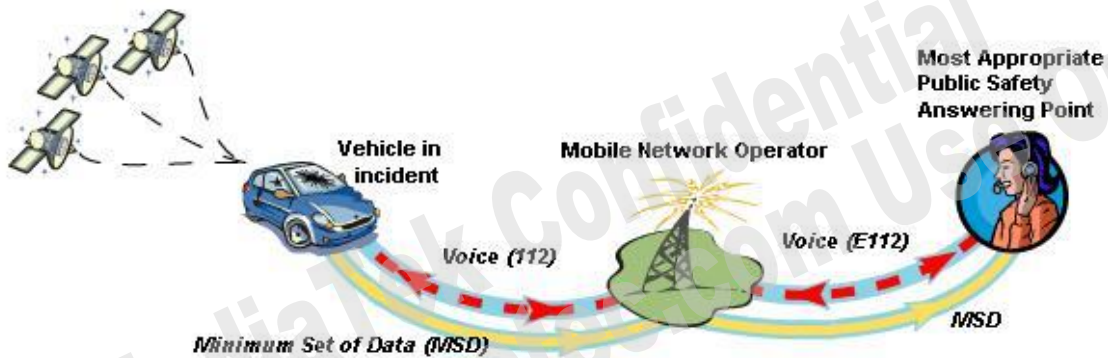


图 1-1. eCall 系统概览

1.1.2 法规介绍

基本上，eCall 规范有两种类型，一种是针对调制解调器协议（如 3GPP 规范），另一种是针对高层应用规范（如 EN 规范）。通常，MTK 将涵盖调制解调器协议部分，并为客户提供 SDK，以便将高层应用规范与自己的应用程序集成。以下是对这些不同规范的简要介绍。有关详细信息，请参阅官方网站上的原始规范。

表 1-1.CS eCall 功能性法规

法规名称	描述
ETSI TS 126.267 (3GPP TS 26.267)	General description
ETSI TS 126.268 (3GPP TS 26.268)	ANSI-C reference code
ETSI TS 127.007 (3GPP TS 27.007)	6.27 Initiate eCall +CECALL
ETSI TS 122.101 (3GPP TS 22.101)	10.7 Transfer of data during emergency calls
ETSI TS 124.008 (3GPP TS 24.008)	4.4.7 eCall inactivity procedure
3GPP TS31.102	SIM related requirement for eCall

法规名称	描述
ETSI TS 102 936-1	5. UMTS network access eCall NAD conformance tests 6.GSM network access eCall NAD conformance tests
EN 16062	eCall high level application requirements (HLAP)
EN 16072	PE eCall Operating Requirements
EN 15722	eCall minimum set of data (MSD)
GOST 33465-2015	Protocols of data exchange between in-vehicle emergency call device/system and emergency response system infrastructure
GOST 33464-2015	In-vehicle emergency call device/system. General technical requirement

表 1-2. CS eCall 测试法规

法规名称	描述
ETSI TS 126.269 (3GPP TS 26.269)	eCall Data Transfer; In-band modem solution; conformance test
ETSI TS134 123-1	13.3 eCall Emergency Call Procedures
ETSI TS 151 010-1	Conformance specification 26.9.6a Structured Calls/eCall
ETSI TS103 412	Pan-European eCall end to end and in-band modem conformance testing; Prose test specification
EN16454	Intelligent transport systems-ESafety-Ecall end to end conformance testing
GOST 33467-2015	Functional test methods of in-vehicle emergency call device/system and data transfer protocols
GOST 33470-2015	Test methods for wireless communication modules of in-vehicle emergency call system
ETSI TS 103 428	eCall HLAP Interoperability Testing

表 1-3. IMS eCall 功能性法规

法规名称	描述
TS24.229	eCall over IMS (NG-eCall) general spec
RFC8147	NG-eCall SIP part spec
TS23.122	NG-eCall eCall only mode
TS24.301	EMM requirement for eCall only mode
TS23.401	eCall only mode
TS24.008	MM requirement for eCall only mode
TS23.216	NG-eCall SRVCC
TS23.167	eCall over IMS (NG-eCall) general spec
TS27.007	AT CMD of eCall
TS31.102	SIM-related requirement for eCall

表 1-4. IMS eCall 测试法规

法规名称	描述
TS34.229-1	SIP layer test spec of NG-eCall
TS36.523	UE conformance test spec of NG-eCall

1.2 eCall 架构

软件架构如图 1-2 所示。目前，MTK 解决方案支持 eCall 设置和带内调制解调器 TX/RX，符合 ETSI/3GPP 标准（绿色矩形），并提供专有 ML 接口调用 eCall RIL 命令。客户需要将 eCall 平台和应用程序（红色矩形）与 eCall ML 接口集成，并使用该产品通过 eCall 测试用例。

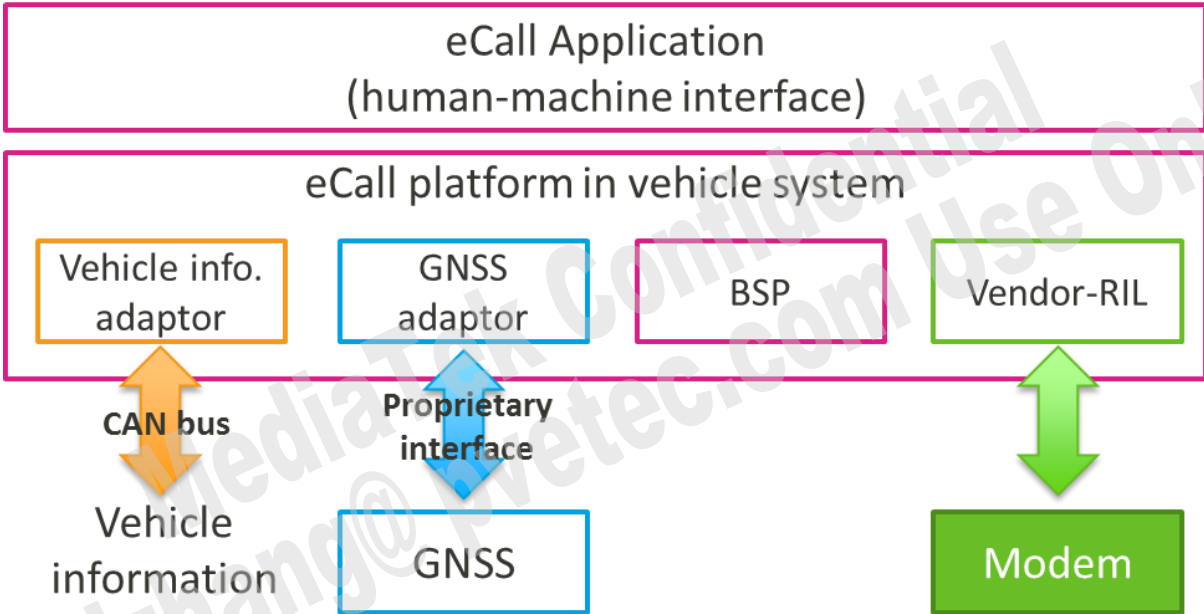


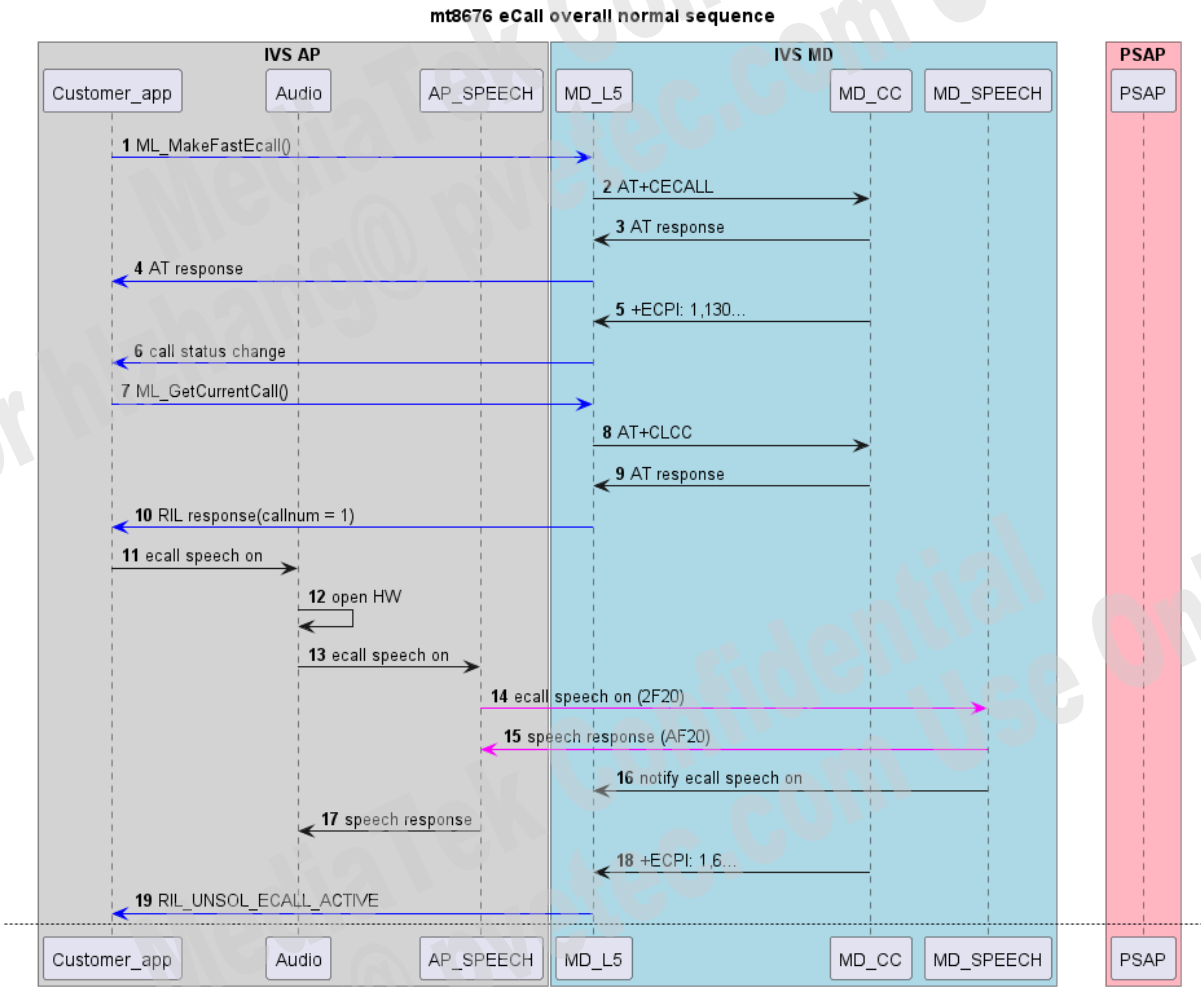
图 1-2. eCall 软件架构

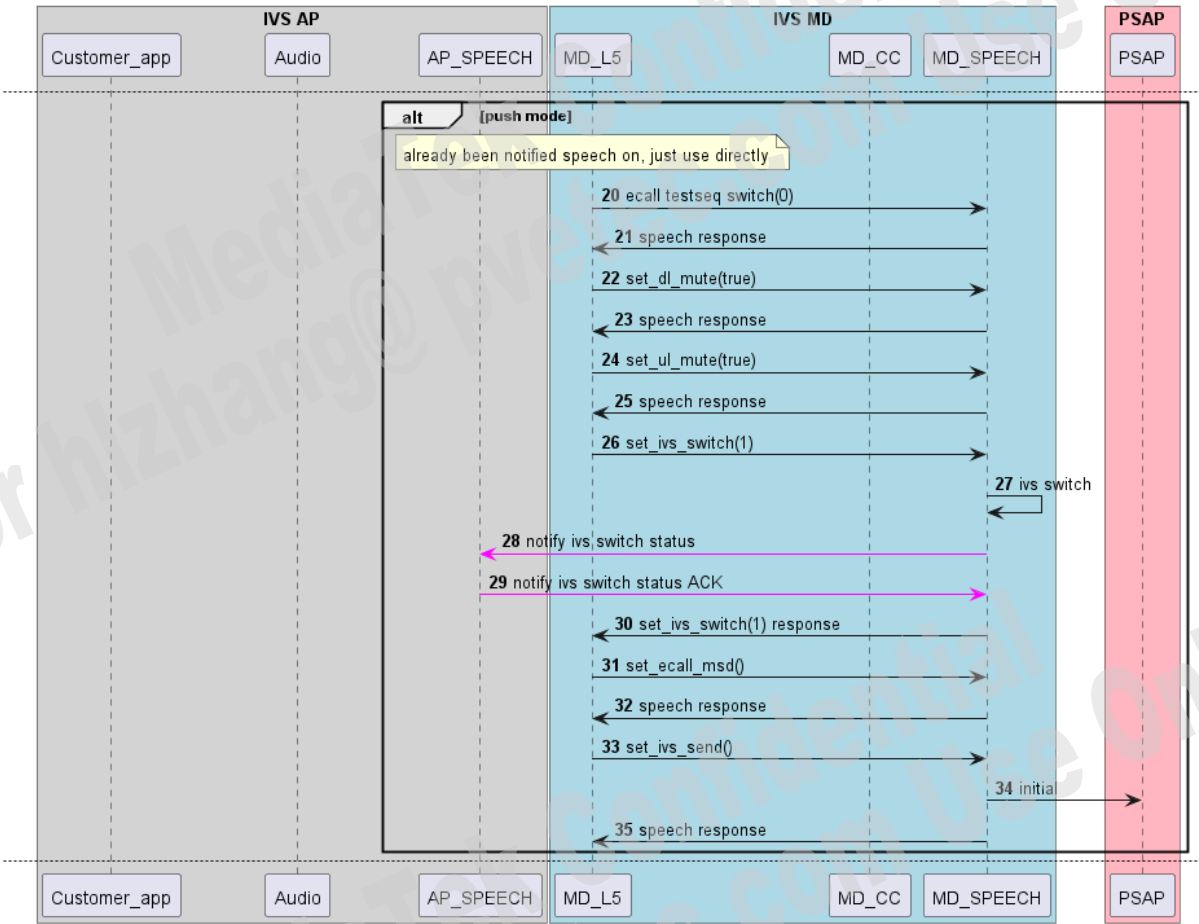
1.2.1 eCall 流程

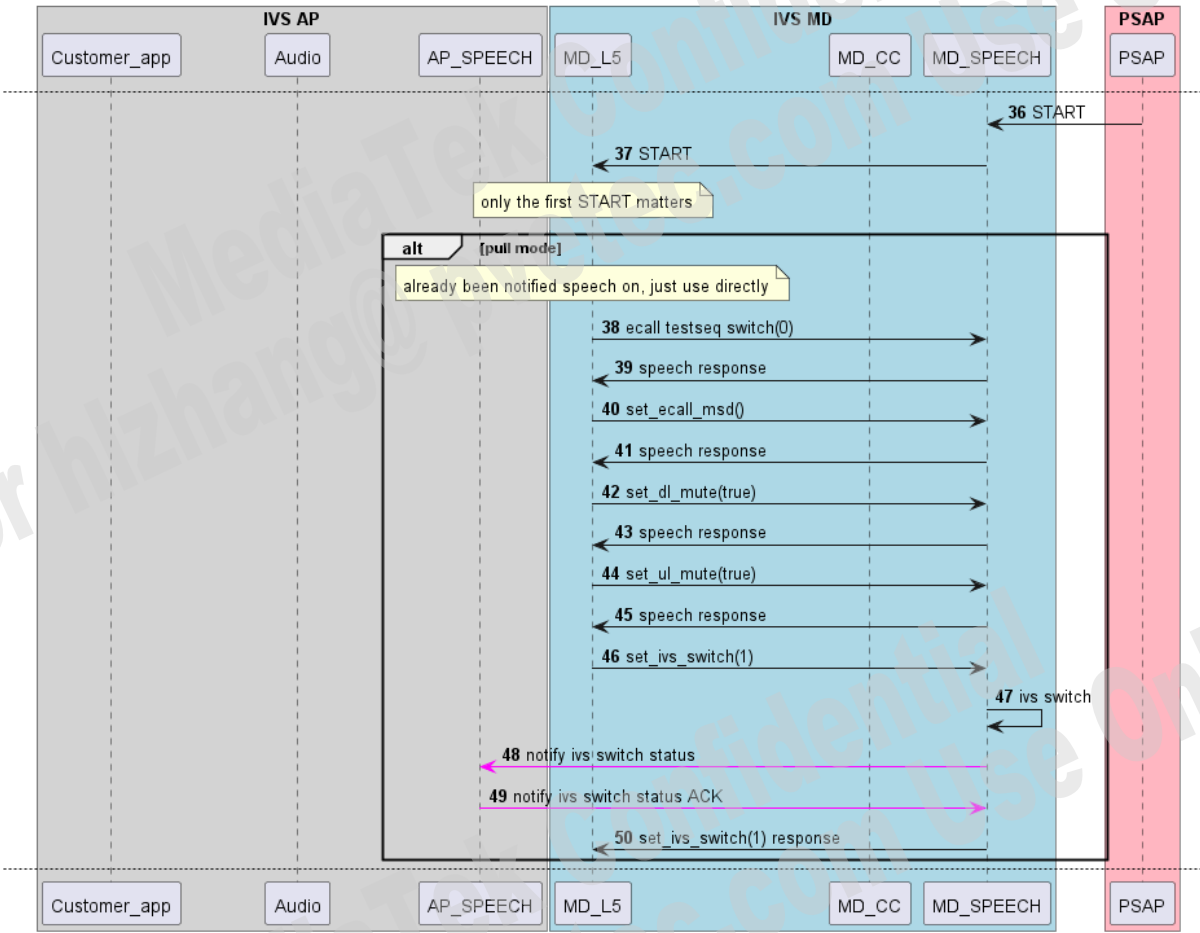
虽然 eCall 场景不同，但关键步骤相同。基本上，客户 APP 需要按照以下步骤处理 eCall 流程：

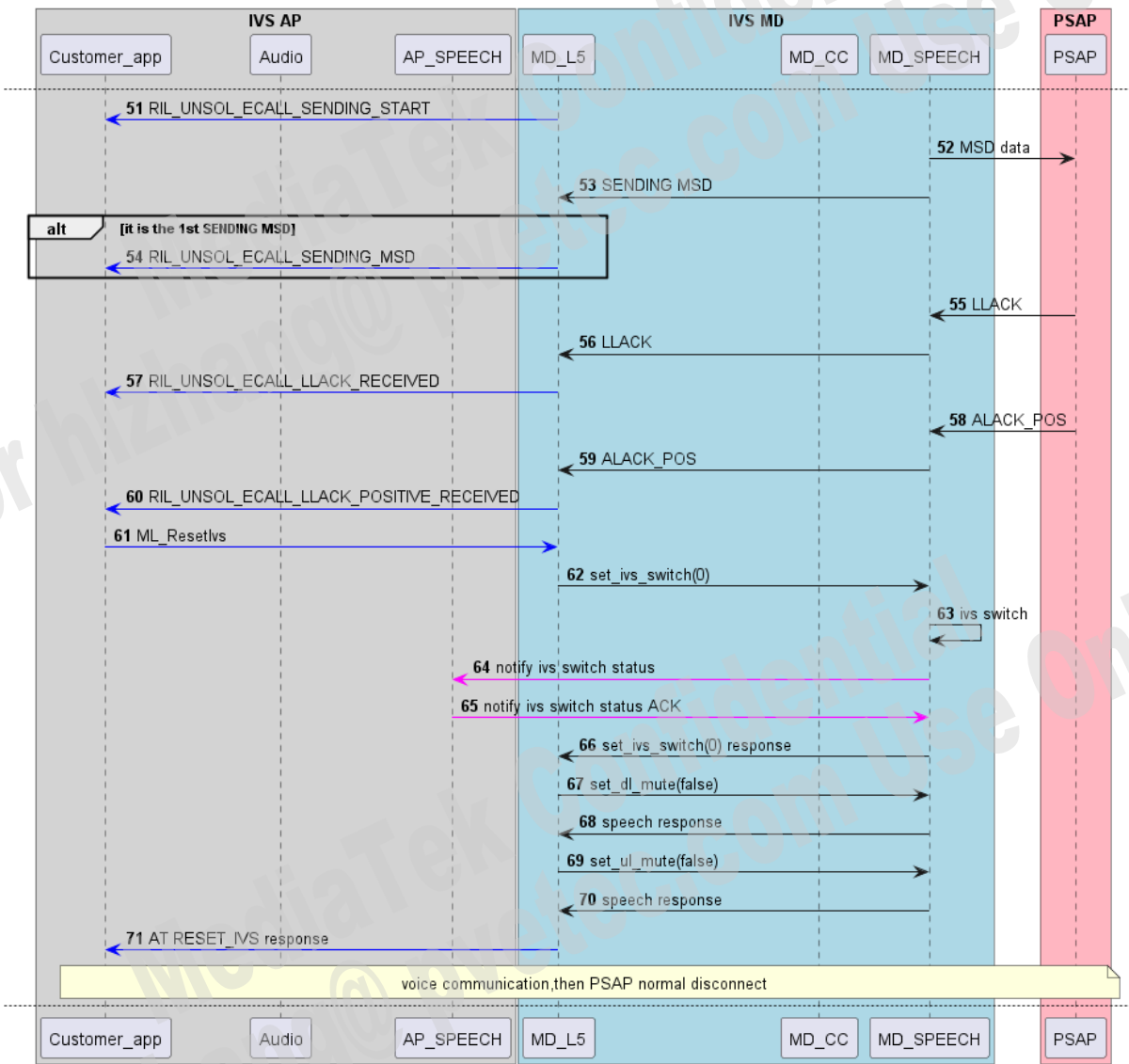
1. 发起 ML_MakeFastEcall;
2. 维护 IVS 侧的 eCall 计时器，该计时器在 EN16062 中指定;
3. 收到 RIL_UNSOL_ECALL_ALACK_POSITIVE_RECEIVED 后发出 RIL_REQUEST_ECALL_RESET_IVS。然后 IVS 和 PSAP 可以恢复语音呼叫连接并相互通话。
4. 收到 RIL_UNSOL_ECALL_DISCONNECTED 或 RIL_UNSOL_ECALL_ABNORMAL_HANGUP 后挂断电话，PSAP 通过此方式指示 IVS 断开呼叫。

以下是 eCall 流程序列的详细描述。









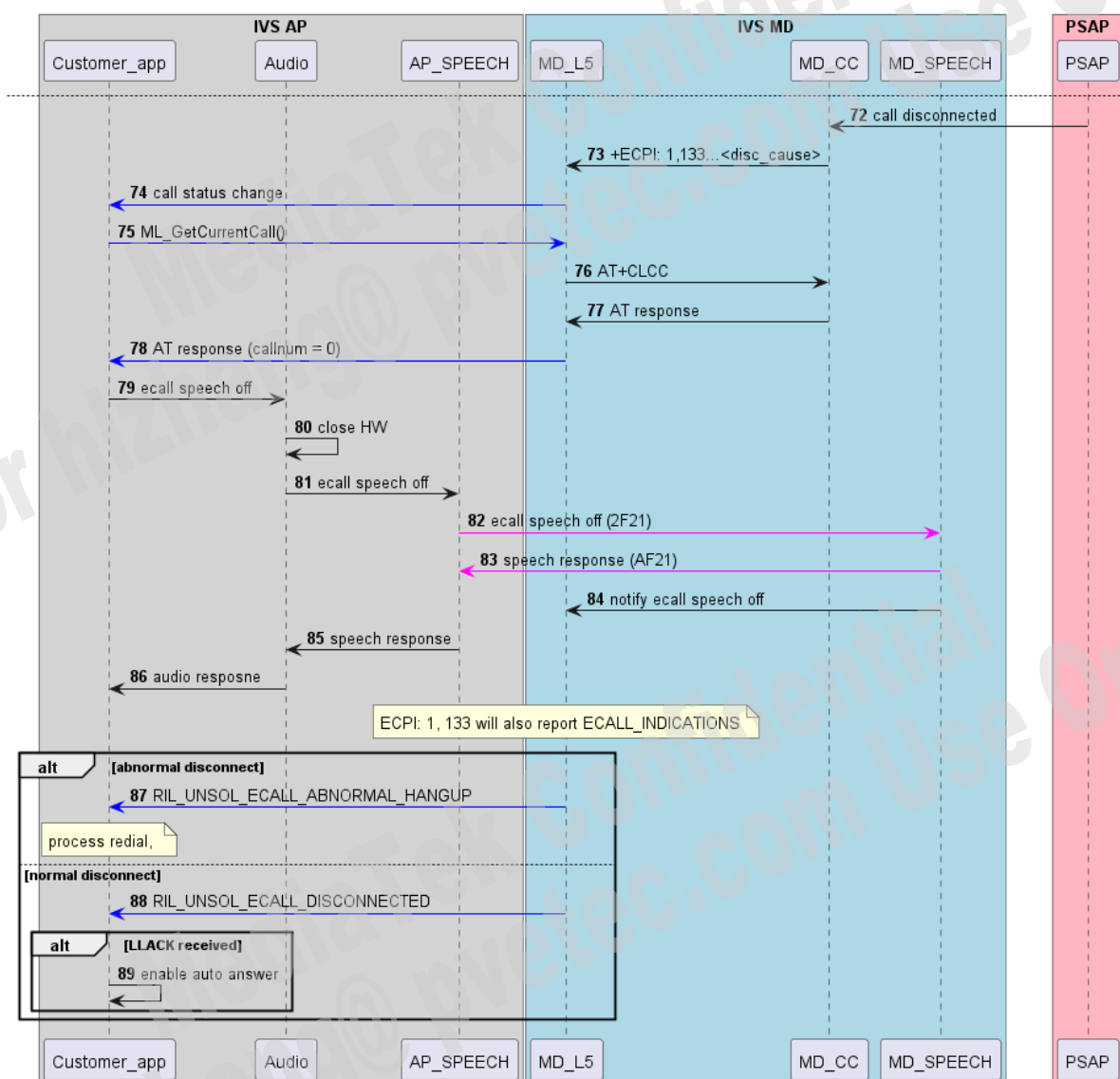


图1-3. eCall 流程

1.2.2 eCall API 使用说明

下列 eCall API 计划在 MT8676 上开发。如果后续开发有更新，将提供更新的文档。

表1-5. eCall 控制接口描述

接口	描述
typedef struct { int32_t call_id; uint32_t length; unsigned char msd_data[ML_ECALL_MSD_MAX_LENGTH]; }ml_ecall_set_msd;	MSD 信息结构体 call_id: 当前电话 index; length: msd data 的长度;

接口	描述
	msd_data: msd data 数据; 传入参数是 byte 格式 (unsigned char), 最大长度 140, 对应的 char 格式最大长度是 280.
<pre>typedef struct { int32_t arg_num; int32_t type; char address[128]; }ml_ecall_set_num;</pre>	Test number/Reconfiguration number 结构体 arg_num: 传入参数个数, 比如传入有效的 type 和 address, 则为 2, 比如只传入有效的 address, 则为 1; type: 传入参数类型, 1 表示传入 URI 类型, 2 表示传入 number 类型; address: Test 或者 Reconfiguration 号码。
<pre>typedef enum { ML_EMER_CAT_MANUAL_ECALL = 1, ML_EMER_CAT_AUTO_ECALL = 2, }ml_ecall_category;</pre>	发生 eCall 类别的结构体
<pre>typedef enum { ML_ECALL_TEST = 1, ML_ECALL_EMERGENCY = 2, ML_ECALL_RECONFIG = 3, }ml_ecall_variant;</pre>	eCall 类型的结构体
<pre>typedef enum{ ML_DOMAIN_AUTO = 0, /* Automatic mode - LTE(IMS), WG(CS), 1x(C2K) */ ML_DOMAIN_CS_ONLY = 1, /* CS domain only - WG(CS) */ ML_DOMAIN_3GPP_ONLY = 2, /* 3GPP only - LTE(IMS), WG(CS) */ ML_DOMAIN_3GPP2 = 3, /* 3GPP2 only - 1x(C2K) */ ML_DOMAIN_IMS_1xCS = 4, /* IMS and 1x CS only - LTE(IMS), 1x(C2K) */ ML_DOMAIN_CS_1x = 5, /* WG CS and 1x CS only - WG(CS), 1x(C2K) */ ML_DOMAIN_IMS_ONLY = 6, /* only IMS call allowed */ }ml_ecall_domain;</pre>	eCall 通话的网络制式结构体
<pre>typedef struct{ ml_ecall_category ecall_cat;</pre>	请求拨出 eCall 时传入结构体参数定义

接口	描述
<pre> ml_ecall_variant ecall_variant; char address[20]; uint32_t length; unsigned char msd_data[ML_ECALL_MSD_MAX_LENGTH]; ml_ecall_domain domain; }ml_ecall_req_msg; </pre>	
<pre> typedef struct{ int32_t data1; int32_t data2; int32_t data3; int32_t data4; }ml_ecall_pri; </pre>	eCall 优先级参数结构体. data1>data2>data3>data4. 传入参数应为 1 2 3 4, 含义分别为: 1: 客户设定的 eCall URI; 2: USIM 保存的 eCall URI 3: 客户设定的 eCall 号码 4: USIM 保存的 eCall 号码
<pre> typedef enum{ E_ML_ECALL_SENDING_START = 1, E_ML_ECALL_SENDING_MSD = 2, E_ML_ECALL_LACK_RECEIVED = 3, E_ML_ECALL_ALACK_POSITIVE_RECEIVED = 4, E_ML_ECALL_ALACK_CLEARDOWN_RECEIVED = 5, E_ML_ECALL_DIALING = 9, E_ML_ECALL_ALERTING = 10, E_ML_ECALL_ACTIVE = 11, E_ML_ECALL_DISCONNECTED = 12, E_ML_ECALL_IMS_ACTIVE = 13, E_ML_ECALL_IMS_DISCONNECTED = 14, E_ML_ECALL_ABNORMAL_HANGUP=15, E_ML_ECALL_IMS_MSD_ACK = 20, E_ML_ECALL_IMS_UPDATE_MSD = 21, E_ML_ECALL_IMS_IN_BAND_TRANSFER = 22, E_ML_ECALL_IMS_MSD_NACK = 23, E_ML_ECALL_IMS_SRVCC = 24, E_ML_ECALL_ONLY_DEREGISTRATION = 31, E_ML_ECALL_MAY_DEREGISTER = 32, E_ML_ECALL_PSAP_CALLBACK_START = 40, E_ML_ECALL_PSAP_CALLBACK_IMS_UPDATE_MSD = 41, </pre>	eCall 上报状态的消息类型枚举定义

接口	描述
<pre> E_ML_ECALL_T2_TIMEOUT = 52, E_ML_ECALL_T5_TIMEOUT = 55, E_ML_ECALL_T6_TIMEOUT = 56, E_ML_ECALL_T7_TIMEOUT = 57, E_ML_ECALL_UNSPECIFIED = 0xffff, }ML_ECall_Indication; </pre>	
<pre> typedef struct{ ML_ECall_Indication ind; int call_id; } ML_ECALL_IND_T; </pre>	eCall 上报状态的消息结构体
<pre> typedef void (*ML_ECALL_MSGCB_T)(ML_ECALL_IND_T *pvsMsg); </pre>	eCall 上报状态消息的回调函数
<pre> int32_t ML_EcallIndicationInit(ML_ECALL_MSGCB_T cb_func); </pre>	注册 eCall 状态变化 callback 函数，当 eCall 状态发生变化时回调 cb_func 函数 参数: 输入: ML_ECALL_MSGCB_T cb_func
<pre> int32_t ML_ResetIvs(void); </pre>	重置 modem 的 eCall 状态。 重连 audio
<pre> int32_t ML_SetMSD(ml_ecall_set_msd* msd); </pre>	设置 eCall 的 msd 数据 参数: 输入: ml_ecall_set_msd* msd
<pre> int32_t ML_SetTestNumber(ml_ecall_set_num* test_num); </pre>	设置 test number 或者 URI 参数: 输入: ml_ecall_set_num* test_num
<pre> int32_t ML_SetReconfNumber(ml_ecall_set_num* reconf_num); </pre>	设置 reconfig number 或者 URI. 参数: 输入: ml_ecall_set_num* test_num
<pre> int32_t ML_MakeFastEcall(ml_ecall_req_msg* msg); </pre>	拨出 eCall 同时传入 msd 数据 参数: 输入: ml_ecall_req_msg* msg
<pre> int32_t ML_SetEmsdpri(ml_ecall_pri* pri); </pre>	设置 eCall 优先级类别. 默认优先级是 "1>3>2>4" 参数:

接口	描述
	输入: ml_ecall_pri* pri

1.3 eCall 定时器

1.3.1 EU CS eCall 定时器

在 eCall 流程中，EN 16062 定义了 10 个定时器。MTK 在调制解调器端实现了 T2、T3、T5、T6、T7、T10。T1、T9 应由 IVS 客户 APP 处理；T4 和 T8 是 PSAP 定时器，应由 PSAP 应用程序处理。

表 1-6. EN16062 中定义的 eCall 定时器

Name Origin	Description Requirements	Value
T1 IVS	Manually initiated eCall(MIeC) false triggering cancellation period - START: T1 starts as soon as the eCall is manually activated - STOP: T1 stops when Vehicle occupants cancel the manually triggered eCall transaction. - EXPIRY: Upon expiry of T1 the IVS-NAD shall start call setup	
T2 IVS	IVS Call Cleardown Fallback Timer (CCFT) - START: T2 starts as soon as the IVS-NAD starts with call setup - STOP: T2 stops when the IVS-NAD receives a call clear-down indication from the mobile network or a call clear-down message from the PSAP. - EXPIRY: Upon expiry of T2 the IVS-NAD shall clear down the call	3600 s (1h)
T3 IVS	IVS INITIATION signal duration - START: T3 is started as soon as the IVS-NAD starts sending the INITIATION signal - STOP: T3 stops when the IVS-NAD receives a SEND MSD signal from the PSAP, at which time the IVS-NAD shall stop sending the INITIATION signal. - EXPIRY: Upon expiry of T3 the IVS-NAD shall stop sending the INITIATION signal	2 s
T4 PSAP	PSAP wait for INITIATION signal period - START: T4 starts as soon as the PSAP eCall modem has answered the call - STOP: T4 stops when the PSAP eCall modem detects an INITIATION signal send by the IVS. - EXPIRY: Upon expiry of T4, the PSAP eCall modem shall route the call to a PSAP operator	5 s
T5 IVS	IVS wait for SEND MSD period START: T5 starts as soon as the IVS-NAD received notification that the call is first answered	5 s

Name Origin	Description Requirements	Value
	- STOP: T5 stops when the IVS-NAD detects a SEND MSD signal sent by the PSAP. - EXPIRY: Upon expiry of T5 the IVS-NAD shall reconnect the IVS audio system and terminate eCall specific behavior(i.e. it shall not proceed with the sending of MSD data) until requested to do otherwise.	
T6 IVS	IVS wait for AL-ACK period - START: T6 starts as soon as the IVS-NAD has received LL-ACK - STOP: T6 stops when the IVS-NAD receives an AL-ACK message - EXPIRY: Upon expiry of T6, the IVS-NAD shall mark the transfer of the MSD as unsuccessful and reconnect the IVS audio system and terminate eCall specific behavior until requested to do otherwise	5 s
T7 IVS	IVS MSD maximum transmission time - START: T7 starts as soon as the IVS-NAD starts sending the MSD data - STOP: T7 stops when the IVS-NAD receives an LL-ACK message - EXPIRY: Upon expiry of T7, the IVS-NAD shall mark the transfer of the MSD as unsuccessful and reconnect the IVS audio system and terminate eCall specific behavior until requested to do otherwise	20 s
T8 PSAP	PSAP MSD maximum reception time - START: T8 starts as soon as the PSAP starts sending the SEND MSD signal - STOP: T8 stops when the PSAP eCall modem receives a valid MSD (reception being acknowledged by sending an LL-ACK) - EXPIRY: Upon expiry of T8, the PSAP eCall modem shall route the call to a PSAP operator	20 s
T9 IVS	IVS NAD minimum network registration period - START: T9 starts as soon as the IVS-NAD clears down a call, or gets notified that a call has been cleared down in accordance with EN 16072 Clause 7.17.3 - STOP: T9 is uninterruptable; until T9 expires the IVS-NAD shall remain registered on the serving network, and remain available to receive calls from the PSAP and rescue workers - EXPIRY: Upon expiry of T9, the IVS-NAD may deregister from the serving network (see T10)	3600 s
T10 IVS	IVS NAD network 'Deregistration Fallback Timer' (DFT) - START: T10 starts as soon as the IVS-NAD clears down a call, or gets notified that a call has been cleared down - STOP: T10 stops if the IVS-NAD receives or makes a new call - EXPIRY: Upon expiry of T10, the IVS-NAD shall deregister itself from the serving network	12 h

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